Baltimore Ravens Data Scientist Application Project — Executive Summary

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For this project, I wanted to examine the pulling ability of offensive linemen on designed run plays. Specifically, I wanted to look at how offensive guards pull on gap scheme (i.e. power, counter, and trap) run plays. The central question is this: can we measure how well a guard pulls by using tracking data, instead of simply relying on game film and the “eye test”?

Using tracking data from the 2025 NFL Big Data Bowl, I identified design run plays where at least one lineman is pulling. I then filtered the plays to include only gap scheme run plays, and isolated the pulling guard from each of these plays. Then, I calculated three metrics to measure various aspects of a guard’s pulling ability: path efficiency (how closely the actual pull path resembles the “ideal” straight-line path), initial burst (how quickly a guard gets off the ball in the first half-second), and time saved above expected (how much quicker a guard arrives at their blocking target than expected). For the latter metric, in order to arrive at an “expected” figure, I compared each guard to how an average guard would perform on the same pull.

Of the three metrics I calculated, path efficiency bears the closest resemblance to what coaches and scouts would observe on game film. The best pulling guards waste little to no motion on their pulling paths, meaning that they stay very close to the theoretical straight-line path between their initial set position and their blocking target. This is something that can be identified on film, and is also taken into account here. Most of the guards in the data have very little separation between their path efficiencies, with the average figure being above 96%. This means NFL-level guards are near-uniformly excellent on being efficient on their pulling paths.

Initial burst measures the acceleration of pulling guards right after the snap, and it measures a different skill than path efficiency, as some guards can be explosive but not path-efficient (or vice versa). Still, the average guard in the data reached an initial burst of 1.75 yards per seconds squared over the first half-second of a pulling play, which is a very strong figure. As for time saved above expected, very little separated the guards in the data (the gap being only 0.27 seconds between the fastest and the slowest), so time taken on pulls is not a strong differentiator. Nonetheless, the best performers on this metric were Joel Bitonio, Wyatt Teller, and Brandon Scherff (all known for run blocking), which suggests that it is sound.

These metrics are not meant to replace film study and grading, but rather to serve as an additional tool. Scouts can verify what they observe on film (e.g. a guard’s quickness) with measurements derived from objective tracking data, making their work easier and more reliable. They can also use these metrics to compare prospects, giving them a complementary and more direct method instead of relying solely on game film from college.

There are several limitations to my findings. Firstly, the available data only span the first nine weeks of the 2022 NFL season, making the sample size small; as a result, some guards have very few pulls, so rankings based on my metrics should be treated cautiously. I isolated each guard on their own, so team scheme tendencies and teammates were not taken into account. Furthermore, my metrics measure a guard’s ability to get to their blocking target and not their blocking ability, so an elite blocking guard may not necessarily rank highly here.

However, the creation and validation of my metrics suggest that tracking data can be used to measure the pulling ability of guards, something that was previously judged only on film. This gives coaches and scouts a new method for evaluating offensive linemen performance.